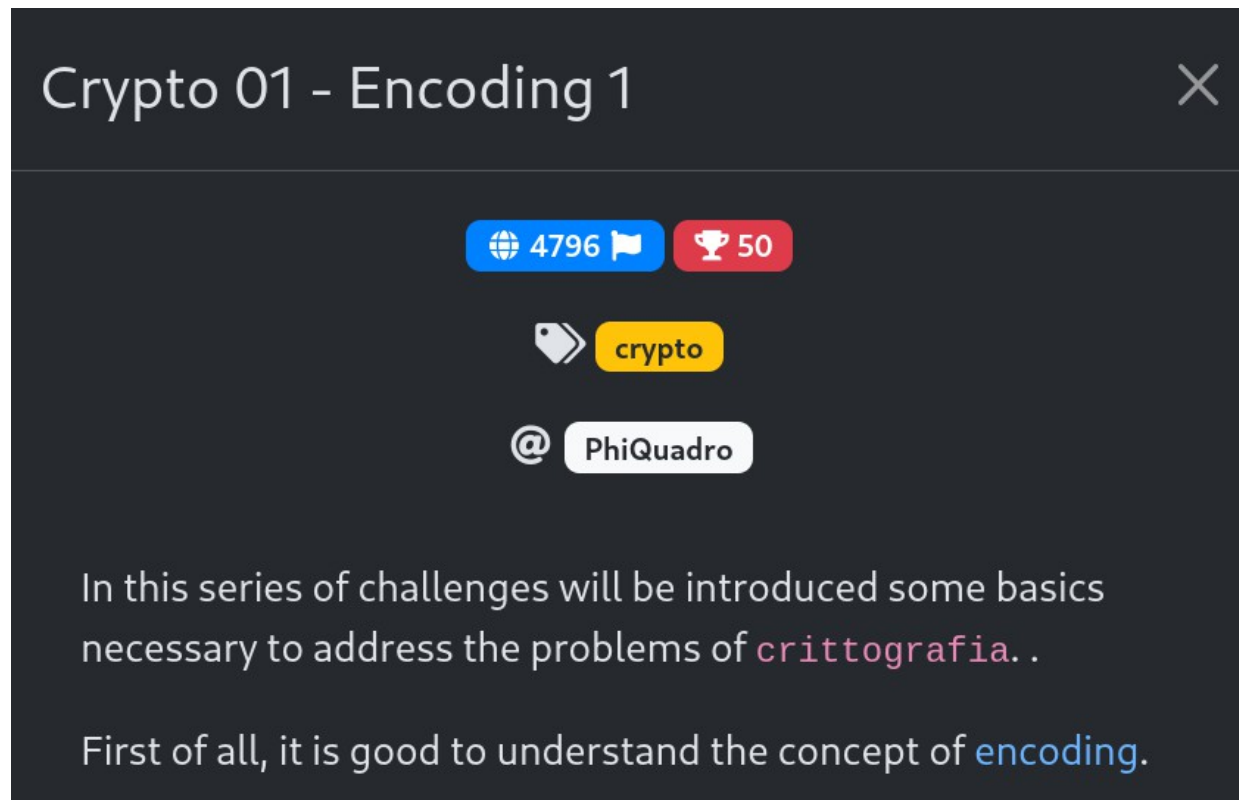


OLlcyber Introduction

Crypto 01 – Encoding 1



In this challenge, we're asked to understand the concept of encoding to start learning cryptography

Crypto 01 – Encoding 1

What is Encoding?

Decimal 65 →
Hexadecimal 41 →
Binary 1000001 →
ASCII A

Encoding is the process of transforming data from one form to another, usually with the goal of storage or transmission using specific systems

Crypto 01 – Encoding 1

What is Encoding?

Decimal 65 →
Hexadecimal 41 →
Binary 1000001 →
ASCII A

Encoding is not cryptography, because the goal of encoding is not to transform information to keep it secret, but it's still useful to illustrate how data can be transformed

Crypto 01 – Encoding 1

ASCII Encoding

Decimal 65 →
Hexadecimal 41 →
Binary 1000001 →
ASCII A

ASCII is an encoding method that is used to transform letters, numbers, symbols, and other characters to code points, which map to specific numbers

Crypto 01 – Encoding 1

ASCII Encoding

The flag of this challenge can be obtained by converting (*decoding*) of these numbers in the corresponding ASCII characters.

```
[102, 108, 97, 103, 123, 117, 103, 104, 95, 78,  
117, 109, 66, 51, 114, 53, 95, 52, 49, 114, 51, 52,  
100, 121, 125]
```

This challenge requires us to convert a list of decimal numbers to ASCII characters, which is encoding

Crypto 01 – Encoding 1

ASCII Encoding

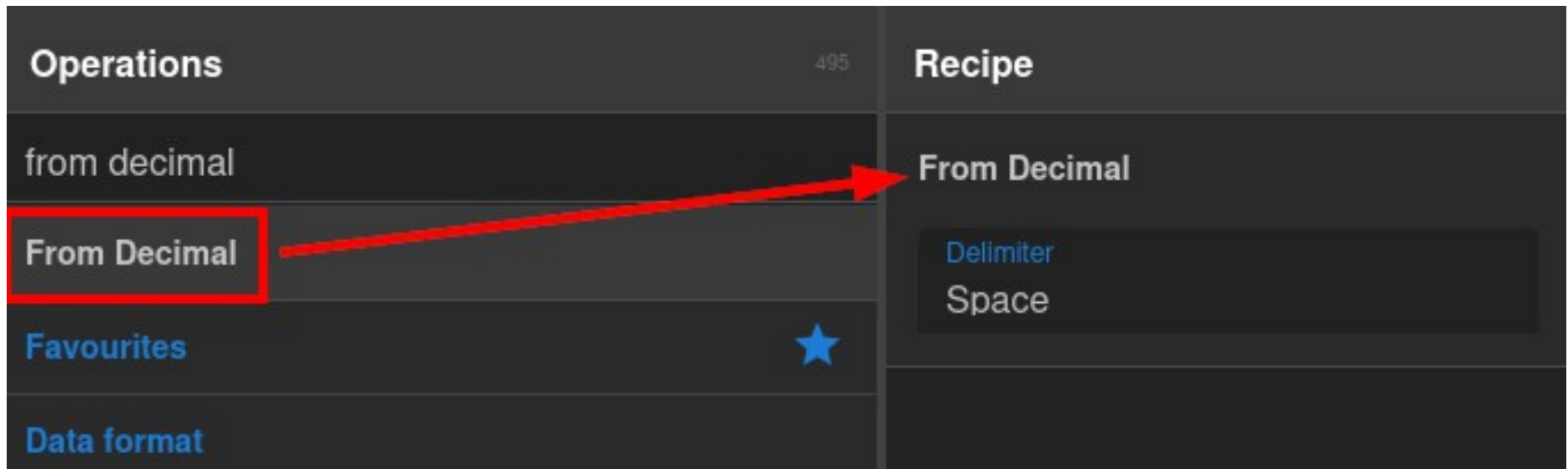
The flag of this challenge can be obtained by converting (*decoding*) of these numbers in the corresponding ASCII characters.

```
[102, 108, 97, 103, 123, 117, 103, 104, 95, 78,  
117, 109, 66, 51, 114, 53, 95, 52, 49, 114, 51, 52,  
100, 121, 125]
```

This challenge requires us to convert a list of decimal numbers to ASCII characters, which is encoding

Crypto 01 – Encoding 1

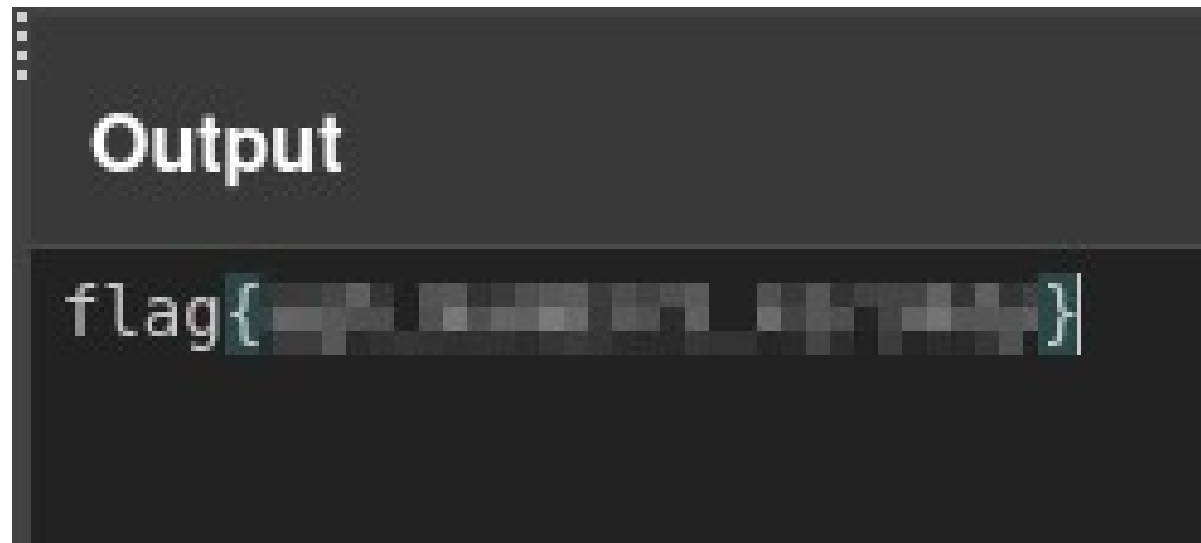
CyberChef Solve



Then, in the `Operations` window, search for the `From Decimal` operation, then drag that selection into the `Recipe` window

Crypto 01 – Encoding 1

CyberChef Solve

A screenshot of the CyberChef application's 'Output' window. The window has a dark background with a light-colored title bar that says 'Output'. Below the title bar, there is a single line of text: 'flag{[REDACTED]}' where the characters inside the curly braces are obscured by a pixelated pattern, representing a redacted flag.

```
Output
```

```
flag{[REDACTED]}
```

The operation will convert the decimal numbers to their ASCII character equivalents, which we can see in the `Output` window

Crypto 01 – Encoding 1

Linux Solve

```
echo '102, 108...' | tr ", " "\n" |  
awk '{ printf "%c", $1 }'
```

We can solve this challenge using Linux commands by using the `echo`, `tr`, and `awk` commands

Crypto 01 – Encoding 1

Linux Solve

```
echo '102, 108...' | tr ", " "\n" |  
awk '{ printf "%c", $1 }'
```

First we `echo` out the list of numbers for use with the other commands

Crypto 01 – Encoding 1

Linux Solve

```
echo '102, 108...' | tr ", " "\n" |  
awk '{ printf "%c", $1 }'
```

We then pipe that output into the `tr` command, replacing the commas with line breaks

Crypto 01 – Encoding 1

Linux Solve

```
echo '102, 108...' | tr ", " "\n" |  
awk '{ printf "%c", $1 }'
```

Finally, we use the `awk` command, which transforms the numbers into its ASCII character, printing them out next to each other

Crypto 01 – Encoding 1

Python Solve

```
num_list = [102, 108, 97, 103, 123, 117,  
  
flag_string = ""  
  
for i in num_list:  
    flag_string = flag_string + chr(i)  
  
print(flag_string)
```

To solve this challenge with Python, we can use the whole list (including the square brackets) as a list variable

Crypto 01 – Encoding 1

Python Solve

```
num_list = [102, 108, 97, 103, 123, 117,  
  
flag_string = ""  
  
for i in num_list:  
    flag_string = flag_string + chr(i)  
  
print(flag_string)
```

Then create an empty string (flag_string), which we will later populate

Crypto 01 – Encoding 1

Python Solve

```
num_list = [102, 108, 97, 103, 123, 117,  
  
flag_string = ""  
  
for i in num_list:  
    flag_string = flag_string + chr(i)  
  
print(flag_string)
```

Then run a for loop over the number list, which adds the ASCII encoded character of each number in the list (using the `chr` function) to the `flag_string` variable

Crypto 01 – Encoding 1

Python Solve

```
num_list = [102, 108, 97, 103, 123, 117,  
  
flag_string = ""  
  
for i in num_list:  
    flag_string = flag_string + chr(i)  
  
print(flag_string)
```

Finally, we print out the populated `flag_string` variable, which contains the flag